

Tharun Sridhar

Backend Engineer | Bengaluru, India

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SUMMARY

Backend engineer (B.Tech CSE, May 2026) who ships production Python APIs with Django REST Framework and FastAPI, backed by PostgreSQL data modeling, JWT authentication, Redis caching, Celery workers, Docker, and CI/CD pipelines. Built the same inventory-management system on both stacks to benchmark them head-to-head.

TECHNICAL SKILLS

Languages: Python, SQL, Java

Backend: Django, Django REST Framework, FastAPI, REST API Design, JWT Authentication, SQLAlchemy 2.0, Alembic, Celery, pytest

Databases: PostgreSQL 16, MySQL, Data Modeling, Indexing, Transaction Isolation, Row-Level Locking

DevOps & Cloud: Docker, Docker Compose, GitHub Actions (CI/CD), Redis, Gunicorn, Railway, AWS, Git, Linux

AI / ML: PyTorch, TensorFlow, Scikit-learn, LLM Integration, OpenCV, NumPy

Frontend: React.js, HTML, CSS

EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science | CGPA: 8.0/10

Vellore, India

May 2026

PROJECTS

Inventra (*Team Lead*) | *Django, DRF, FastAPI, PostgreSQL 16, SQLAlchemy 2.0, Alembic, JWT, Docker* [GitHub](#)

- Led a team building a role-based inventory backend (Admin/Manager/Employee); personally built JWT auth (access + revocable refresh tokens) and the REST API over a versioned, indexed PostgreSQL 16 schema.
- Made stock mutations atomic and idempotent (append-only ledger) guarded by `SELECT ... FOR UPDATE` row-level locking in fixed order; regression-tested against a real concurrency race.
- Shipped 51 endpoints across 13 modules with search, filtering, and pagination; containerized and CI-tested against PostgreSQL, then reimplemented the identical spec in FastAPI + SQLAlchemy to benchmark both stacks.

PhotoShare API | *Django, DRF, FastAPI, PostgreSQL, Redis, Celery, JWT, ImageKit CDN, Docker* [GitHub](#)

- Built a photo/video sharing API with a full auth lifecycle (registration, verification, forgot/reset-password); reset tokens bind to the password hash so old tokens invalidate on change.
- Streamed multipart uploads to ImageKit CDN, added Redis cache-aside caching on feed/post-detail reads with write invalidation, and moved thumbnail generation to a retry-safe Celery worker with backoff.
- Enforced ownership-based authorization (403 on non-owner deletes); packaged API, worker, Postgres, and Redis via Docker Compose, with a parallel Django + DRF implementation of the same routes.

NeuroScan AI (*Team Lead*) | *Python, FastAPI, TensorFlow, OpenCV, Groq LLM* [GitHub](#)

- Directed a team on a brain-tumor MRI pipeline; personally trained a four-model classification ensemble and built the FastAPI serving layer, including an Attention U-Net stage reaching a 0.88 Dice score.
- Integrated Groq LLM report generation and PDF export via FastAPI, gating results into Accepted/Caution/Specialist-Review tiers using a Grad-CAM-vs-segmentation reliability index, covered by four pytest suites.

Finvoro | *Django, DRF, PostgreSQL, JWT, pytest* [GitHub](#)

- Built a personal finance API (Django REST Framework) for multi-account tracking, categorized transactions, monthly budgets, and spending reports, with balances computed live rather than cached.
- Enforced per-user data ownership on every endpoint via JWT (django-rest-framework-simplejwt), versioned under `/api/v1/`, and covered with a pytest + pytest-django suite plus Swagger/ReDoc docs.

CERTIFICATIONS

AWS Solutions Architect Associate (SAA-C03) – Exam Prep Course

Packt (Coursera)

ACHIEVEMENTS

5★ HackerRank: Problem Solving, Java, Python, SQL | hackerrank.com/profile/tharunsridhar

162 LeetCode problems solved (73.75% acceptance) | leetcode.com/u/Tharunsridhar